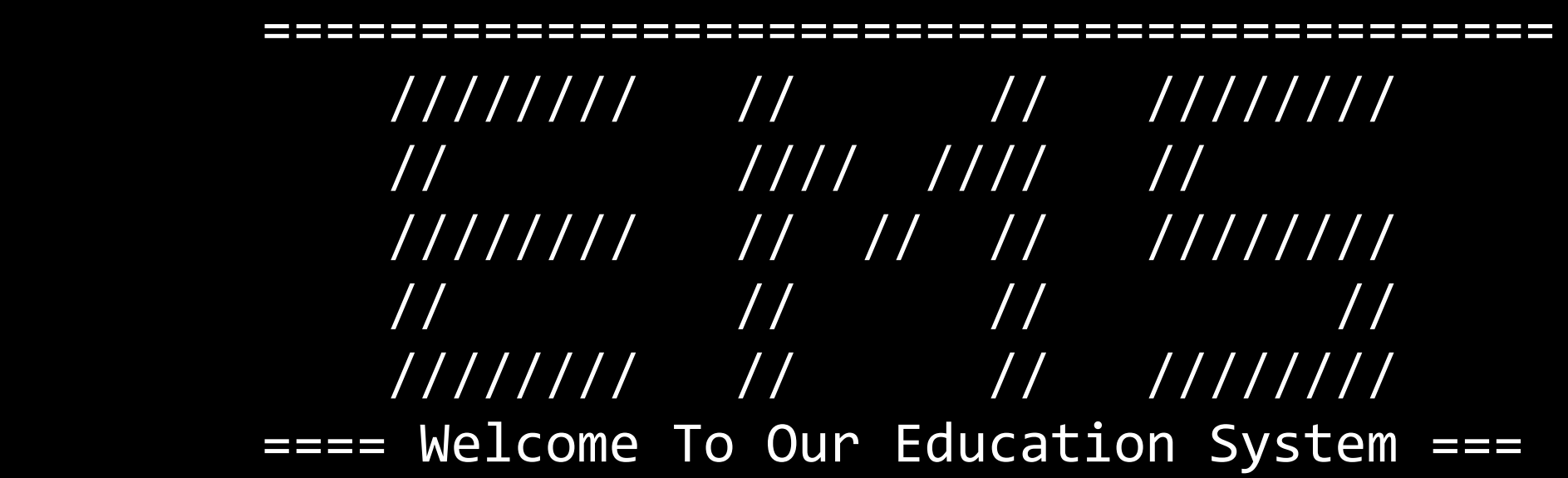



```
C:\EMS> list team_members
[1] Steven Ashraf
[2] Mohanad Mohsen
[3] Shady Adel
[4] Fares Osama
[5] Joyce Phillip


Press Any Key To Continue .....
```



C:\EMS> project info

"Our project is an integrated Educational Management System (EMS) developed as a C# Console Application. It aims to simulate smart learning platforms by organizing and managing the educational process among all key stakeholders: Faculty Members (Doctors), Teaching Assistants (TAs), and Students. The system relies on a Role-Based Access Control (RBAC) approach, where each user category gets its own dedicated dashboard and features."

> What does the app do?

A role-based Educational Management System

Built as a C# console application: no GUI, only menus.

[1] USERS & AUTH

> Students sign up, everyone logs in

> Each role gets its own dashboard

[2] COURSES

> Doctors create courses and add TAs

> Students register and view courses

[3] ASSIGNMENTS & GRADING

> Doctors post assignments with deadlines

> Students submit, doctors grade + comment

[4] COMMUNICATION

> Doctors publish course announcements

> Students receive notifications

> The main menu

```
========================================
    ////////   //      //   ////////
    //         ////  ////   //
    ////////   //  //  //   ////////
    //         //      //         //
    ////////   //      //   ////////
==== Welcome To Our Education System ===

1. Login
2. Sign-up ( Student )
3. Exit
Select an Option :
```

1. Login

Asks for UserID + password, calls AuthService.Login() and opens the dashboard that matches the role.

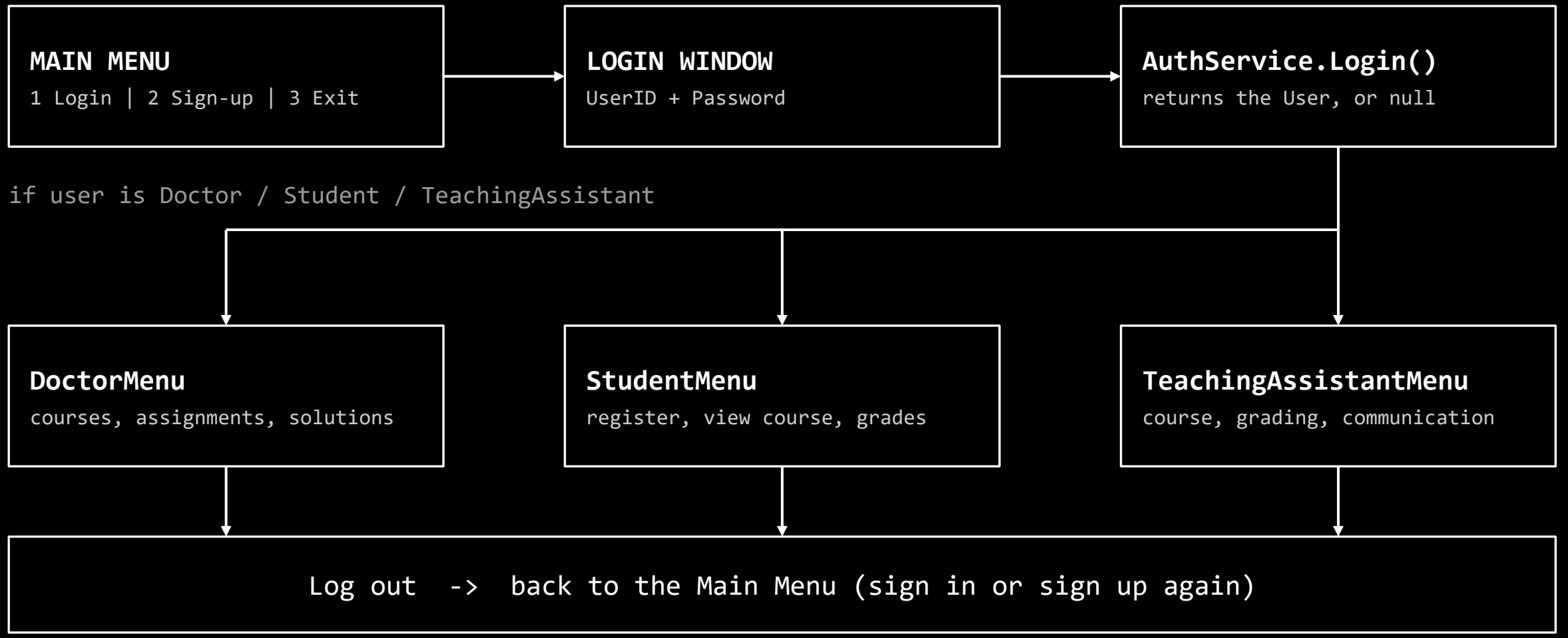
2. Sign-up (Student)

Asks for full name and a Gmail / Hotmail email, and repeats until the data is valid.

3. Exit

Ends the loop and prints "Goodbye !".

> How the system flows



> Project structure

UI

MainMenu · DoctorMenu · StudentMenu
TeachingAssistantMenu

↓

SERVICES

AuthService · CourseService · AssignmentService
GradingService · CommunicationService

↓

REPOSITORIES

User · Course · Assignment · AssignmentSolution

↓

MODELS

User (Doctor, Student, TeachingAssistant)
Course · Assignment · AssignmentSolution

INTERFACES

Every service and repository has an interface (IAuthService, ICourseRepository ...).

Services get their repositories and validator through the constructor.

Menus talk only to services, never to data directly.

> Roles in the system

DOCTOR

> Creates courses and adds TAs

> Posts assignments with max grade + due date

> Views solutions, sets grades + comments

> Publishes course announcements

STUDENT

> Signs up with name + email

> Registers in courses

> Views courses + assignment status

> Submits one solution per assignment

> Views grades and notifications

TEACHING ASSISTANT

> Assigned to courses

> Menu gets course, assignment, grading and communication services

> Who built what

[1] Users & Auth	Shady Adel
[2] Courses	Fares Osama
[3] Assignments	Mohanad Mohsen
[4] Grading	Steven Ashraf
[5] Communication	Joyce Phillip

> Module 1: Users & Auth

owner: Shady Adel

RESPONSIBILITIES

> Create accounts and manage users

> Login and change passwords

> Validate user data

> Manage user roles and permissions

> Show success / error messages in the console

UNDER THE HOOD

> Login(userId, password) returns a User or null

> SignUp(fullName, email) returns the new UserId

> Sign-up: name cannot be empty, email must be Gmail or Hotmail, repeats until valid

> After login the user's role decides the menu: Doctor, Student or TA

> Module 2: Courses

owner: Fares Osama

RESPONSIBILITIES

> Display and create courses

> Manage course info and link courses to doctors

> Manage student registration

> Assign Teaching Assistants

> Display course details in the console

UNDER THE HOOD

> Create / Update / Delete course, GetCoursesByDoctor

> Validation: name, code, doctor id, MaxStudents > 0. Code saved in UPPERCASE

> RegisterStudent blocks duplicates and checks the registration start / end dates

> DeleteCourse cleans up its assignments, solutions and links to students + TAs

> Module 3: Assignments

owner: Mohanad Mohsen

RESPONSIBILITIES

> Display assignments for courses

> Create assignments: title, description, max grade, due date

> Let students submit solutions

> Validate submissions and deadlines

> Save and display submission status

UNDER THE HOOD

> CreateAssignment: title up to 200 chars, MaxGrade > 0, due date in the future, course must belong to that doctor

> SubmitSolution: assignment not expired, student registered in the course, one submission per student

> Solution text: not empty, max 10,000 chars

> Module 4: Grading

owner: Steven Ashraf

RESPONSIBILITIES

> Display student submissions

> Assign grades and add comments

> Validate grades against the assignment's max grade

> Save grades and make results available to students

UNDER THE HOOD

> ViewSolution(id) and GetSolutionsByAssignment(id)

> SetGrade(id, grade): rejects negative grades and grades above MaxGrade

> SetComment(id, comment): rejects empty text, trims it

> ViewGradesReport(studentId) returns that student's solutions with grades

> Module 5: Communication

owner: Joyce Phillip

RESPONSIBILITIES

> Display messages and announcements

> Publish course announcements

> Target announcements to the appropriate users

> Save and display messages in the console

UNDER THE HOOD

> CreateAnnouncement(doctorId, courseId, title, message): the course must belong to the doctor, title + message required

> SendNotification(studentId, message): only to Student users

> DisplayNotifications(studentId) shows only that student's messages

> Everything is kept in memory lists

> The end

```
C:\EMS> exit
```

Thank you for using the system , Goodbye !

Any questions ?_